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DOES PLANT SPECIES CO-OCCURRENCE INFLUENCE SOIL MITE DIVERSITY?

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Abstract. Few studies have considered whether plant taxa can be used as predictors of belowground faunal diversity in natural ecosystems. We examined soil mite (Acari) diversity beneath six grass species at the Konza Prairie Biological Station, Kansas, USA. We tested the hypotheses that soil mite species richness, abundance, and taxonomic diversity are greater (1) beneath grasses in dicultures (different species) compared to monocultures (same species), (2) beneath grasses of higher resource quality (lower C:N) compared to lower resource quality, and (3) beneath heterogeneous mixes of grasses (C_3 and C_4 grasses growing together) compared to homogeneous mixes (C_3 or C_4 grasses) using natural occurrences of plant species as treatments. This study is the first to examine the interaction between above- and belowground diversity in a natural setting with species-level resolution of a hyper-diverse taxon.

Our results indicate that grasses in diculture supported a more species and phylogenetically rich soil mite fauna than was observed for monocultures and that this relationship was significant at depth but not in the upper soil horizon. We noted that mite species richness was not linearly related to grass species richness, which suggests that simple extrapolations of soil faunal diversity based on plant species inventories may underestimate the richness of associated soil mite communities. The distribution of mite size classes in dicultures was considerably different than those for monocultures. There was no difference in soil mite richness between grass combinations of differing resource quality, or resource heterogeneity.

Key words: Acari; biodiversity; functional groups; microarthropods; microhabitat; mites; overyield; resource heterogeneity; resource quality; soil ecosystem; species richness.

INTRODUCTION

Knowledge of how, or even if, aboveground and belowground diversity are related is important to our understanding of biodiversity, conservation, ecology, and the services ecosystems provide to society (Hooper et al. 2000, Wall et al. 2004, Wardle et al. 2004). We are experiencing unprecedented global change be it climate, land use, or species extinctions; yet there is a critical lack of information as to what the implications of these changes are to soil organisms and the ecological processes they impact (Wall et al. 2004). Part of the problem is that the overwhelming diversity and abundance of soil organisms requires many experts and intense labor for study in all but the most simplified laboratory experiments (Giller 1996, Lawton et al. 1998). Plant richness can be a useful indicator of aboveground animal diversity at local scales (Currie 1991, Simonson et al. 2001); however, whether plant species richness is a determinant of belowground species richness has not been tested in a natural ecosystem. The few studies addressing this question were laboratory experiments (De Deyn et al. 2003), field-plot manipulations (Wardle et al. 1999, 2004, De Deyn et al. 2004), studies of human-created disturbance (Korthals et al. 2001, St. John et al. 2002, Huhta and Niemi 2003), and/or used higher taxonomic or functional groupings (Wardle et al. 1999, Porazinska et al. 2003). Soil mites (Acari) were examined rarely, at high taxonomic levels, and generally showed no consistent patterns relating to aboveground diversity (Hooper et al. 2000, Wardle et al. 2004). We investigated the relationship between above- and belowground diversity using grass species and soil mites, a hyper-diverse taxon, in uncultivated tall grass prairie.

Soils communities are primarily detrital-based (Hunt et al. 1987, Coleman and Crossley 1996). Their function and composition are implicitly linked to the quantity and quality of resources from plant matter aboveground and to the roots of these plants (Bardgett and Cook 1998, Jones et al. 1998). Laboratory and field manipulations demonstrate that plant richness may be positively related to the diversity of soil litter quality or litter types that enter the detrital food web (Anderson 1978). This resource heterogeneity may support a greater diversity of soil fauna through both the provision of food substrate and additional microhabitats

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(Anderson 1978, Berg et al. 1998, Hansen 2000, Clapperton et al. 2002). Aboveground plant richness and diversity may be considered a surrogate for, or determinant of, belowground faunal richness and diversity through mechanisms of altering aspects of resource quantity, quality, and heterogeneity.

We tested this hypothesis using grass species' characteristics and the communities of soil mites associated with them, at the Konza Prairie Biological Research Station (KPBS; 39.1° N, 94.6° E). The KPBS is part of the last remaining tracts of unplowed tallgrass prairie in the United States and among the National Science Foundation's network of Long Term Ecological Research (LTER) sites.

Mites are the most abundant members of soil arthropod communities, accounting for as much as 95% of all individuals (Behan et al. 1978, Seastedt 1984). Mites affect decomposition by feeding on microbes and detritus, as predators, omnivores, and as plant parasites (Walter 1987, Kethley 1990, Siepel and De Ruiter-Dijkman 1993, Walter and Proctor 1999). Their global taxonomic richness is estimated to exceed 1×10^6 species, with the majority of species thought to be soil dwellers (Walter and Proctor 1999). As with many soil mesofauna, feeding habits and functional roles for most species are often inferred from studies of closely related species (Marshall et al. 1987, Wall et al. 2001).

We tested three hypotheses that soil mite richness and taxonomic distinctness (a measure of how unrelated species are to each other, Δ^*; Warwick and Clarke 1995) are determined by (1) grass species richness, or co-occurrence (one vs. two), (2) grass resource quality, based on photosynthetic pathway, and (3) grass resource heterogeneity based on mixtures of photosynthetic pathways. We measured mite species richness from soil sampled between two individuals of six predetermined species of grasses, three C_3 and three C_4 (Table 1). The photosynthetic pathways C_3 and C_4 represent quality of plant resources, including roots, with C_3 plants characterized by lower C:N and lignin content than C_4 plants (Wedin and Tilman 1990, Tilman et al. 1997). We measured several soil quality covariates including C:N of soil, roots and shoots, and soil textures from these grass combinations. We determined if juxtaposed grass species in diculture results in a nonadditive increase (overyielding) of soil mite richness compared with two grass species in monoculture.

TABLE 1. Sampling design for the collection of soil cores and their subsequent use in testing the hypotheses for grass richness (H1, monoculture [1] vs. diculture [2]), quality (H2, low [C_4C_4] vs. high [C_3C_3]), and heterogeneity (H3, low [C_3C_3 or C_4C_4] vs. high [C_3C_4]) effects on soil mite richness.

Grass combination†	Grass		
	Richness	Quality	Heterogeneity
AA	1	C_4	*1*
BB	1	C_4	*1*
CC	1	C_3	*1*
DD	1	C_3	*1*
EE	*1*	C_4	*1*
FF	*1*	C_3	*1*
AB	2	*C_4*	1
AC	2	*C_3C_4*	2
AD	2	*C_3C_4*	2
BC	2	*C_3C_4*	2
BD	2	*C_3C_4*	2
CD	2	*C_3*	1

Note: Italicized text indicates samples not used for given analyses.

† (A) Big bluestem (*A. gerardii*), C_4; (B) little bluestem (*S. scoparium*), C_4; (C) Scribner's panicum (*D. oligosanthes*), C_3; (D) Kentucky bluegrass (*P. pratensis*), C_3; (E) prairie dropseed (*S. heterolepis*), C_4; and (F) prairie junegrass (*K. nitida*), C_3.

MATERIALS AND METHODS

Study site description

The KPBS has >300 documented species of vascular plants in the area and is dominated by big bluestem (*Andropogon gerardii* Vitm.) and little bluestem (*Schizachyrium scoparium* (Michx.) Nash), both C_4, perennial grasses (Freeman 1998). We chose a watershed (KPBS designation ''4B'') that is experimentally burned every four years. The last burn occurred the year previous to this study (May 1998). Its intermediate fire frequency is thought to mimic natural wildfire conditions and it supports a diverse mixture of both C_3 and C_4 grasses (Collins and Steinauer 1998). For more descriptions of the soil characteristics and site see a parallel study, Porazinska et al. (2003).

Sampling design

On 18 May 1999 four replicate blocks (~20 × 20 m) were established ~100 m apart on the upper side of watershed 4B's slope. Grass species pairs in 12 different combinations (Table 1) were marked with a pin flag and replicated five times within each block. Selection of grass pairs was nonrandom in order to minimize the influence of nontarget plant species' roots; the distance between pairs was never >10 cm and no shoots of nontarget plant species were within a 20 cm diameter. The selected grass species were all native or naturalized at Konza: (A) big bluestem (C_4), (B) little bluestem (C_4), (C) Scribner's panicum (*Dichanthelium oligosanthes* (Schult.) Gould) (C_3), (D) Kentucky bluegrass (*Poa pratensis* L.) (C_3), (E) prairie dropseed (*Sporobolus heterolepis* (A. Gray) A. Gray) (C_4), and (F) prairie junegrass (*Koeleria nitida* Nutt.) (C_3).

Soils for every plant species combination, from one of the randomly selected five replicate pin flag locations within each block, were sampled with a soil corer (4.8 cm in diameter) to two depths (0–5 cm, 5–10 cm). Two more core samples were each taken from another of the replicate set of pin flag locations for plant and soil chemical and physical analyses. Additionally, soil samples from beneath the remaining replicate pin flag lo-

cations were collected to analyze other members of the soil community (Porazinska et al. 2003).

Table 1 shows how the 12 grass species combinations were used to test the three hypotheses of this experiment. Grass species richness was defined as the number of plant species (one or two; monoculture or diculture) within 10 cm of the soil core. Where both individuals of grass were the same species and had a C_4 photosynthetic pathway (C_4C_4), resource quality was considered to be low compared to grass pairs of the C_3 photosynthetic pathway (C_3C_3). Where two grasses of different species and had the same photosynthetic pathway (C_3C_3 or C_4C_4), resource heterogeneity was considered to be low or homogeneous compared with heterogeneous photosynthetic grass pairs (C_3C_4).

Soil mites

Soil samples were carefully wrapped in aluminum foil, placed in coolers and transported to the KPBS soil laboratory within two hours of sampling. There, microarthropods were actively extracted from the soil cores using a high-gradient modified Tullgren method (Crossley and Blair 1991) into 95% ethanol and stored until identified. All sampled mites from the three suborders Mesostigmata, Oribatida, and Prostigmata were enumerated and identified, in order of increasing taxonomic resolution, to superfamily, family, genus, and species based on manuals from the Acarology Summer Program (Ohio State University, Columbus, Ohio, USA), numerous published descriptions and texts (Krantz 1978, Dindal 1990), and the assistance of mite systematists (E. E. Lindquist [Agriculture and Agri-Food Canada] and D. E. Walter [University of Alberta]). Only adult mites were used in statistical analyses since many groups have polymorphic immature stages that cannot be reliably assigned to a species. Adult mites were assigned to the seven functional groups: comminuting microbivores–detritivores (grazers and browsers), piercing–sucking microbivores, plant parasites, nematophages, arthropod predators, generalist predators, and omnivores (Anderson 1975, Walter and Proctor 1999).

The size of each mite species was determined from average length and widths obtained by measuring random subsets of up to 10 adult individuals per mite species for total body length and maximum width (exclusive of mouthparts and appendages).

Mite specimen data were managed using Biota, a fully relational database software application suited for biodiversity studies (Colwell 1997). Voucher specimens have been deposited at the Kansas State University Museum of Entomological and Prairie Arthropod Research (KSU-MEPAR, Manhattan, Kansas, USA). Raw data including species lists are available on the Konga Prairie Biological Station web site).[4]

[4] ⟨http://www.konza.ksu.edu/biodiversity/ksbdrpnet.html⟩

Plant and soil covariates

See Porazinska et al. (2003) for procedures used to measure microbial carbon (C) and nitrogen (N), soil C and N mineralization rates, root C and N, total soil moisture, C, N, and organic matter (SOM), and soil texture (percentage sand, silt, and clay).

Data analysis

Statistics were calculated with SAS version 8 (SAS 1999). Stepwise regressions (maximum R^2) of measured soil and plant covariates against our treatment designations (e.g., C_3 vs. C_4) and mite richness were performed to determine which covariates should be used in ANCOVAs. Richness was calculated for each core, for each horizon, and compared using repeated-measures ANOVA in PROC MIXED where horizon was repeated within core. Block (1–4) was entered as a random factor while grass richness (1 vs. 2), quality (C_3C_3 vs. C_4C_4), and heterogeneity (C_3C_3 or C_4C_4 vs. C_3C_4) were entered as fixed factors. The distribution of mite width classes between mono- and diculture samples was compared using a chi-square test.

To test for overyielding of mite richness with grass richness, composite samples were created from the unions of monoculture samples and compared in a repeated-measures ANOVA, to the union of two appropriate diculture samples:

$$XX \cup YY < XY \cup XY$$

where X and Y represent two different species of grasses (e.g., big bluestem (A) and little bluestem (B)), and XX, YY, and XY represent the mite fauna recorded from soil cores taken from between two individuals of these species of grass (e.g., AA, BB, and AB). This was repeated for every possible combination of samples producing unique composite samples.

Results

Mites were, as expected, numerous (6961 mites total from all cores; average density of 84 000 mites/m^2 at 0–10 cm depth) and species rich (162 species within three soil mite taxonomic suborders Mesostigmata, Oribatida, and Prostigmata). All species of grass had similar associated mite richness ($F_{5,33} = 1.59$, $P = 0.19$) and mite abundance ($F_{5,33} = 0.89$, $P = 0.50$). See Porazinska et al. (2003) for results of root and soil chemical analyses.

H1: grass species richness influences on soil mite richness

Grass and mite species richness were positively related at 5–10 cm depth (Fig. 1). Grass species richness and the richness of mites at taxonomic resolutions lower than species were positively related when considering the 0–10 cm soil horizon (Fig. 2). The richness of mite families and superfamilies was positively related to grass richness for the entire soil sample from

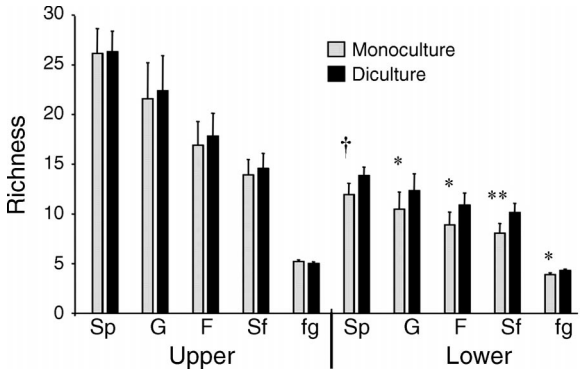

FIG. 1. Taxonomic and functional richness of soil mites from grass mono- and dicultures, split by soil horizon, at the Konza Prairie Biological Station, Kansas, USA. Mites are grouped according to resolution of identification on the *x*-axis (Sp, species; G, genus; F, family; Sf, superfamily; fg, functional group). Significant differences between mono- and dicultures are indicated by the number of asterisks (†$P < 0.1$; *$P < 0.05$; **$P < 0.01$; ***$P < 0.001$). Error bars represent +SE.

0–10 cm depth (Fig. 2). Functional group richness ($F_{1,35} = 4.10$, $P = 0.05$) was positively related to grass richness only in the lower soil horizon (Fig. 1). Mites beneath dicultures were more taxonomically distinct (Δ*; less related) than in monocultures ($F_{1,35} = 13.71$, $P < 0.001$; Fig. 3).

Mite species richness was greater in the lower soil horizon of juxtaposed ($XY \cup XY$) samples compared with similar paired ($XX \cup YY$) samples ($F_{1,101} = 11.96$, $P < 0.001$) indicating that mite richness increased in a nonadditive manner with grass richness at lower soil horizons but not in the upper soil horizons ($F_{1,101} = 0.08$, $P = 0.77$; Fig. 4).

There was an interaction between grass species richness and soil horizon ($F_{1,72} = 4.96$, $P = 0.029$) indicating that total numbers of mites were more equitably distributed in the soil profile of dicultures than monocultures (Fig. 5).

Mite body lengths (l) and widths (w) were highly correlated ($w = 0.713l - 27.218$, $R^2 = 0.88$); thus only widths were used for the discussion of mite sizes. The size class distribution of mites was distinctly different between mono- and dicultures ($X^2_{19} = 132$, $P < 0.001$; Fig. 6). Dicultures supported the highest proportion of mites in the 50–75 μm width class while the greatest proportion of mites in monocultures was in the 100–125 μm width class.

As detailed in Porazinska et al. (2003), C-mineralization rates were higher beneath monocultures (32.1 $\mu g\ C \cdot g^{-1} \cdot d^{-1}$) compared to dicultures (24.4 $\mu g\ C \cdot g^{-1} \cdot d^{-1}$; $F_{1,46} = 6.15$, $P = 0.017$); however, no other soil covariates (microbial C and N, N-mineralization rates, root C and N, total soil moisture, C, N, SOM, and soil texture) were measurably different. Stepwise regression showed significant and positive relationships between initial NO_3 and NH_4 and soil mite richness ($F_{2,53} = 10.74$, $P < 0.0001$). Yet these were not significant when entered as covariates in ANCOVAs and were not considered in further analyses.

H2: grass resource quality influences on soil mite richness

There was no effect of grass resource quality (C_3 vs. C_4) on mite richness at any taxonomic level or functional grouping considered (Fig. 7). Roots of C_3 grass combinations had higher concentrations of root N (0.83 ± 0.14% vs. 0.69 ± 0.11%; $F_{2,45} = 6.982$, $P = 0.002$) and lower C:N (48.7 ± 9.7 vs. 60.4; $F_{2,45} = 5.215$, $P = 0.009$) than C_4 grasses. The single grass species prairie dropseed (E) was responsible for a trend of lower total grass C:N and lower SOM for C_3 compared to C_4 grasses (Porazinska et al. 2003). No other relationships were found between grass resource quality and soil covariates.

H3: grass resource heterogeneity influences on soil mite richness

Grass resource heterogeneity (C_3C_4 vs. C_3C_3 or C_4C_4) had no relationship to mite richness at any taxonomic level or functional grouping considered (Fig. 8). Additionally, no relationship was found between grass resource heterogeneity and any of the measured soil covariates.

DISCUSSION

Our study has shown that the diversity of soil mites in tallgrass prairie is positively related to the richness (one vs. two) of grass species at the scale of individual plants. This relationship also held for mite taxonomic levels above species, their taxonomic distinctness, and functional group richness. Additionally, we found that the co-occurrence of grass species overyielded mite species richness compared to what would be expected from those grasses in monoculture, demonstrating that the grass co-occurrence effect on mite richness is nonadditive. These effects were essentially absent in the topmost soil layer, but significant at depth. The mechanisms for these grass richness effects are not clear

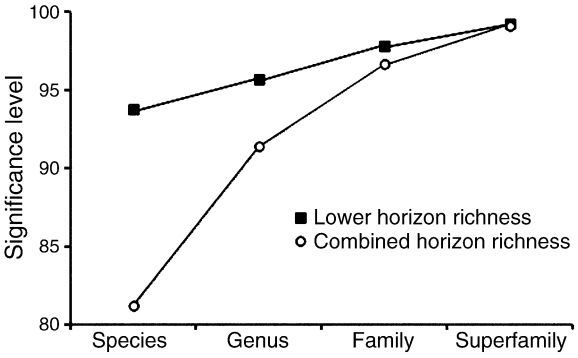

FIG. 2. Significance of grass richness effect (mono- vs. diculture) on mite taxonomic richness; significance level = $100(1 - P)$.

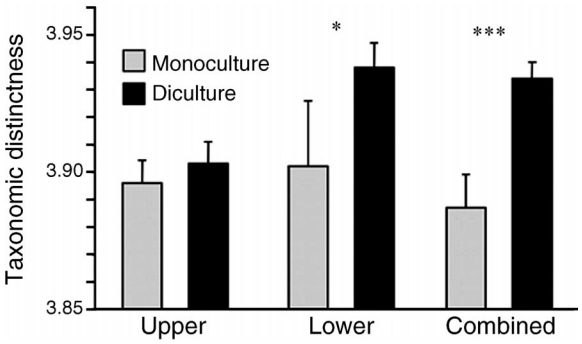

FIG. 3. Taxonomic distinctness of soil mites from grass mono- and dicultures, split by soil horizon, and in combination (i.e., recomposed upper and lower horizon samples). Significance levels shown by asterisks are defined in Fig. 1. Error bars represent +SE.

since neither grass resource quality nor heterogeneity influenced the soil environment or the mite community in a detectable way.

H1: plant species richness influences mite richness

On average there were approximately two more mite species where grass species were in diculture than in monoculture at the KPBS (Figs. 1–4). These findings are consistent with studies showing a modest positive relationship between oribatid mite species richness and the number of dominant tree species of managed forests (Migge et al. 1998), and between nematode and plant species richness in an experimental field study (De Deyn et al. 2004). Most experimental studies, however, show no relationship between aboveground plant diversity and belowground faunal diversity at the species level (Wardle et al. 1999, Korthals et al. 2001, Huhta and Niemi 2003, Salamon et al. 2004). This suggests idiosyncratic relationships between plant and soil faunal richness are possibly due to the specific taxa under consideration and the study design.

The positive relationship between grass and mite species richness was increasingly stronger at higher mite taxonomic levels (superfamily > family > genus > species; Fig. 2), suggesting that mite species are less related to each other beneath dicultures compared to monocultures. Taxonomic distinctness (Δ^*) of the soil mite faunas confirmed that mite species were less related, or more phylogenetically diverse, in diculture compared with monoculture soils (Fig. 3). Thus the richer prairie flora supported both a richer and more phylogenetically diverse soil mite fauna. This result has implications for conservation decisions since given equal species richness it is generally considered a higher conservation priority to conserve for greater phylogenetic diversity (Warwick and Clarke 1995, Faith et al. 2004).

However, soil mite species richness may be underestimated by predictions based solely on grass richness because the relationship between them is nonadditive (Fig. 4). The mechanism for this overyielding of mite richness where grass species were juxtaposed is unknown. Mite richness is known to be positively related to microhabitat diversity (Anderson 1978, Berg et al. 1998, Hansen 2000), and root interspecific interactions affect root architecture, the rhizosphere and hence soil mite habitat (Gordon and Rice 1992). Thus we hypothesize that there was a greater diversity of habitats for mites as a result of root competition where grass species were growing in mixed species combinations compared with monocultures (Figs. 1, 3, 4). Similar results were found in a grassland experiment where the abundance and species richness of Collembola were found to be positively related to fine root biomass and plant functional group diversity (Salamon et al. 2004).

Also, we found that the influence of grass diversity on soil mite diversity was dependent on soil depth (Figs. 1–4). In the lower soil horizon (5–10 cm) mites were consistently richer at all taxonomic levels beneath dicultures compared to monocultures, but differences were not observed in the shallow depths (Fig. 1). This makes sense in light of evidence that prairie grasses in Kansas tend to avoid rooting just below the surface, which may be a drought tolerance or fire avoidance trait (Qian et al. 1997). Any effects of interspecific competition on root architecture, and by extension mite habitat and species richness, will be more pronounced where roots are most active.

Further evidence for this hypothesis was found in the profile of mite abundances (Fig. 5) and body widths (Fig. 6) by depth. Soil mites are affected by soil pore size and distribution (Vreeken-Buijs et al. 1998) and their body widths decrease as pore widths decrease with depth in the soil profile (Curry 1971). In our study the proportion of mite body widths <125 μm was distinctly skewed favoring smaller mites beneath dicultures compared to monocultures (Fig. 6). Also, there was a more

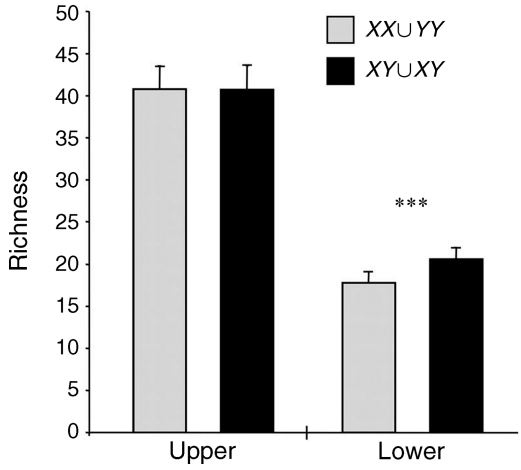

FIG. 4. Species richness of soil mites from combined juxtaposed samples ($XY \cup XY$) compared with combined nonjuxtaposed samples ($XX \cup YY$), split by soil horizon. Error bars represent +SE.

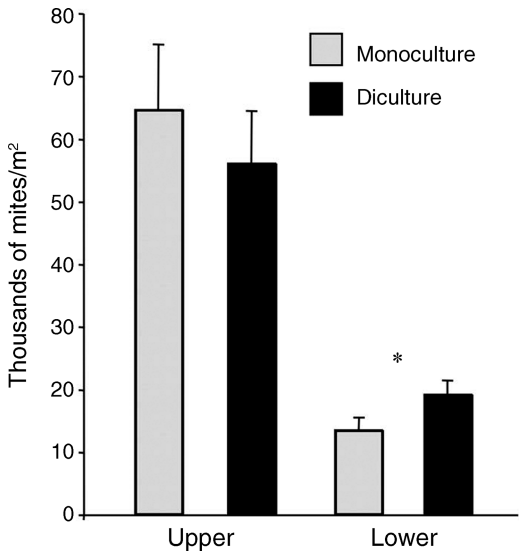

FIG. 5. Abundance (individuals/m^2) of soil mites from grass mono- and dicultures, split by soil horizon. Error bars represent +SE.

equitable distribution of mites between the upper and lower horizons of dicultures compared to monocultures (Fig. 5). These results suggest that the soil associated with dicultures has a more diverse pore structure at depth and a greater abundance of smaller habitable pores than for monocultures; however, this was not measured in our study.

We assigned mite species to seven functional groups in order to compare our results with those for taxonomic groupings as well as other studies. We found that mite functional group richness was positively related to plant species richness as well (Fig. 1). This may suggest that functional groups are reasonable surrogates for mite species; however, immature and adult mites can have different food sources (Walter 1987, Walter and Ikonen 1989, Siepel and De Ruiter-Dijkman 1993) and <20% of the North American soil mite fauna has been described (even fewer have been adequately studied; Marshall et al. 1987, Walter and Proctor 1999) making generalizations tenuous.

In a parallel collaborative study at KPBS the richness of functional groups of nematodes and protozoa was not influenced by the richness of grasses (Porazinska et al. 2003), although other studies have shown nematode and plant diversity to be related at the generic level (Korthals et al. 2001). There are few data on the influence of plant diversity on soil mite or arthropod functional diversity, but some studies indicate that plant identity and functional role (e.g., legumes) are likely to be more important to the relationship than plant diversity per se (Wardle et al. 1999, Setälä 2002, Hedlund et al. 2003, Salamon et al. 2004).

H2: grass resource quality influences on soil mite richness

The diversity of aboveground arthropod communities in grasslands was positively related to the nutrient status and productivity of the plants on which they feed (Siemann 1998). However, we found no effect of grass resource quality (C_3 vs. C_4) on soil mite richness (Fig. 7) at any taxonomic level or functional grouping considered. The lack of grass resource quality effect on soil mites was expected since few of the plant and soil quality metrics measured were different between C_3 or C_4 samples. In a parallel study, Porazinska et al. (2003) found no effect of grass resource quality on the diversity of functional groups of nematodes and protozoa at the KPBS. However, phospholipid fatty acid analysis of the microbial community at Konza showed greater concentrations of bacteria and, in particular, gram-positive bacteria beneath C_3 compared to C_4 grasses (Porazinska et al. 2003). This was likely a response to the lower C:N of the C_3 compared to C_4 grasses being expressed as more labile root exudates and promoting bacterial growth in C_3 soils. It was surprising that this microbial community shift had no influence on the functional richness of higher trophic groups directly dependent on them (protoza and nematodes) but expected that the more trophically distant mites may be unresponsive.

It is also possible that timing of sampling may have obscured quality differences between C_3 and C_4 grasses or that the microbial community changes too rapidly to be able to draw conclusions on its influence for our study. The ''cool season'' C_3 and ''warm season'' C_4 grasses have different phenologies (Hartnett and Fay 1998). We may have observed larger differences between C_3 and C_4 grasses had we sampled several times during the season. Also, when compared with forbs, nitrogen fixers, and woody plants, the differences between C_3 and C_4 grasses are slight. However, Osler and Beattie (2001) found that even though different tree species had strong effects on the soil environment, these influences were not enough to affect the composition of soil mite communities. Thus it appears that soil and litter fauna are relatively unspecific to the

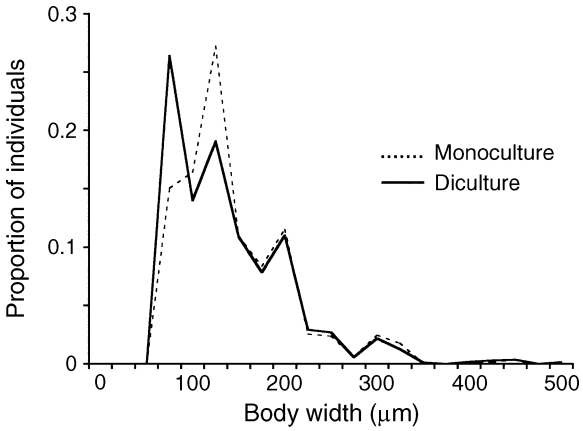

FIG. 6. Proportion of soil mites of increasing body width in mono- and dicultures.

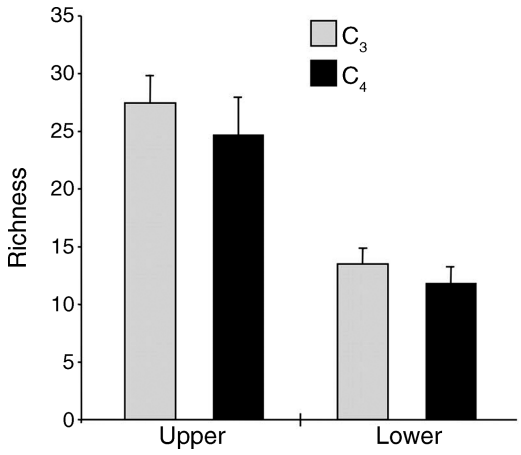

FIG. 7. Soil mite species richness from low-quality (C_4) and high-quality (C_3) grass samples, split by soil horizon. Error bars represent +SE.

plants and their functional aspects when compared to aboveground fauna.

This is a recurring theme in the soil literature; belowground food webs tend not to follow patterns established by observations of aboveground food webs (Setälä 2002). Belowground food webs have characteristics that may override differences in resource quality such as separate energy channels (i.e., root, fungal, and bacterial; Hunt et al. 1987), the commonness of indirect interactions among the fauna, and the high degree of omnivory (Setälä 2002).

H3: grass resource heterogeneity influences on soil mite richness

We found no effect of grass resource heterogeneity (C_3C_4 vs. C_3C_3 or C_4C_4) on soil mite richness (Fig. 8) at any taxonomic level or functional grouping considered. This was unexpected given the evidence that heterogeneous mixtures of grass species (diculture; Fig. 1) did support a richer soil mite fauna compared with monocultures (Figs. 1–4). Similarly, Porazinska et al. (2003) found no influence of grass resource heterogeneity on the functional diversity of protozoan, nematode, or microbial communities. Migge et al. (1998) found no influence of tree stand type (mixed vs. monoculture) on the diversity of soil oribatid mite communities and Maraun and Scheu (2000) found that oribatids were not influenced by the identity or diversity of plant resources entering the soil system. This was interpreted as a function of the generalist detritivore–fungivore life history of most oribatid mites (Maraun et al. 2003).

This is in apparent contrast to studies where experimental manipulation of leaf material in litter bags demonstrated a positive relationship between plant litter diversity and oribatid mite diversity (Hansen and Coleman 1998, Hansen 2000). However, Hansen (2000) concluded that architectural differences between leaf litters created a greater diversity of mite microhabitats in mixed-species litter bags and that this may have been a more important driving mechanism for mite diversity than its influence on microbial and food resources. This parallels our hypothesis that the mechanism for the positive relationship between grasses and soil mite richness was the architectural difference in grass roots within the soil profile, when grasses were growing in competition with other species rather than conspecifics.

Other factors, not measured in our study, may play a more important role in determining soil mite diversity than plant richness or resource quality and heterogeneity. For example Clapperton et al. (2002) found that the abundance and diversity of soil microarthropods was positively related to grassland productivity and hence resource quantity. Currie (1991) showed that at regional scales richness of aboveground animals was highly correlated to potential evapotranspiration, a measure of how much energy is entering the ecosystem, and not plant richness. This suggests that at larger scales than we investigated, patterns in the richness of soil fauna may be similar to those for some aboveground vertebrates. Also potentially important to the diversity of soil mites is the influence and interactions of other soil organisms. Huhta and Niemi (2003) concluded that the influence of earthworms on soil structure and resources was the most significant factor determining the diversity of oribatid mites. Finally, Sulkava and Huhta (1998) found that soil faunal diversity was positively related to the spatial heterogeneity of resources but not the diversity of resources. That is, different resources in patches promoted a richer fauna than the same resource mixed together presumably for reasons of competitive exclusion.

Our results on the positive, nonlinear relationship between soil mite and grass species richness supports the practice of using floral inventories as biodiversity

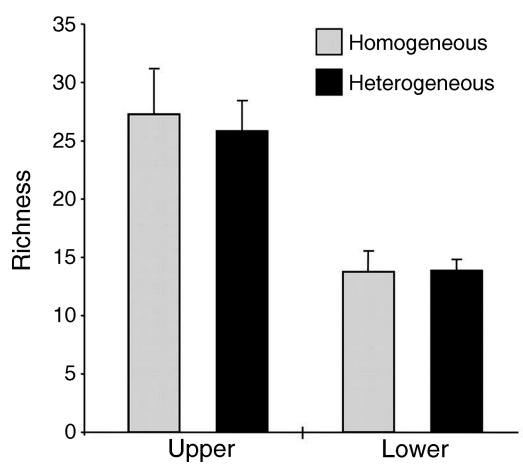

FIG. 8. Soil mite species richness from homo- (C_3C_3 or C_4C_4) and heterogeneous (C_3C_4) grass resource samples, split by soil horizon. Error bars represent +SE.

assessment tools (Simonson et al. 2001) for soil organisms. Although, further investigation is required to determine if the positive relationship between grass and mite species richness holds above the simple one and two grass species combinations presented here and what are the driving mechanisms. Also interesting are the implications of our results for tallgrass prairie soil functioning. Our data suggest that when grass roots compete it affects the soil fauna up to the level of soil mites. This may have a noticeable influence on decomposition and nutrient mineralization processes. By stretching the soil fauna deeper into the soil profile beneath competing grasses more soil volume is becoming biologically active and potentially the spatial arrangement of sites of resource availability, net immobilization, and net mineralization may be altered creating more soil heterogeneity. Any of these has potential to influence overall rates of soil processes.

The relationship between above- and belowground diversity in natural systems is certainly complex and dependent upon the organisms studied as well as the scale of the investigation (Currie 1991, Hooper et al. 2000, Ettema and Wardle 2002, Wardle et al. 2004). Our study adds to this literature by providing data toward understanding the patterns of this aboveground—belowground relationship and in identifying potential mechanisms that warrant experimental study. This knowledge is invaluable for the prediction of soil diversity, its influence on soil processes, and for broader management considerations.

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